

**ET AI
HACKATHON
2.0
PHASE 2:
PROTOTYPE
SUBMISSION**

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Problem Statement 8:

AI for Industrial Knowledge Intelligence:
Unified Asset & Operations Brain

Industrial knowledge is scattered across disconnected systems, making critical information difficult to access when it matters most.

THE CRISIS, IN NUMBERS

35% professionals in asset-intensive industries lose a large share of working hours searching for information, clarifying instructions, or recreating documents that already exist somewhere in the organization.

7-12 operational knowledge is fragmented across disconnected document systems P&IDs in one place, work orders in another, procedures in a third, and inspection records in a fourth.

25% of India's experienced industrial engineers and operators will retire within the next decade, taking undocumented operational knowledge with them, permanently



Knowledge fragmentation is cited as a contributing factor in unplanned downtime events in Indian heavy industry

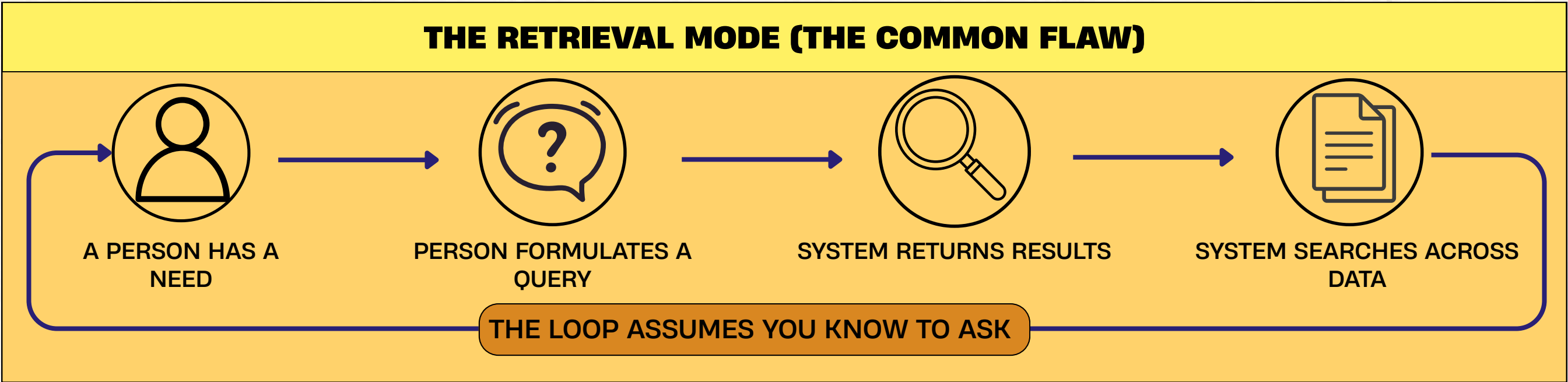
WHY TRADITIONAL SEARCH/RETRIEVAL FAILS?

Document management systems, enterprise search, RAG pipelines, knowledge graphs, predictive maintenance platforms, all share one architectural assumption: **knowledge must be retrieved.**

But the most dangerous knowledge gaps are the ones nobody knows to **query**. A technician doesn't search for a failure history they don't know exists. A planner doesn't search for a regulatory change they don't know happened.

These are not retrieval failures. They are awareness failures. No search engine, however sophisticated, closes an awareness gap.

- Hours lost hunting for information
- Maintenance delays and rework
- Loss of critical tacit knowledge



- Unplanned downtime
- Increased risk of incidents
- Higher operating costs

2. Solution: What is KAIROS ?

THE INVERSION

KAIROS's primary job is not to answer questions. It is to ensure critical knowledge reaches the person who needs it before they know they need it.

THE KAIROS MOMENT:

In Greek, Kairos means the right, decisive moment, as opposed to *chronos*, mere clock time. KAIROS treats operational events (work order creation, permit-to-work issuance, shift handover, alarm acknowledgment, equipment tag-out) as **triggers**, not user actions.

INTERPRET, INFER, PRIORITIZE

Each trigger is interpreted against the full governed knowledge of the organization. The system asks: *what does the person about to do this work almost certainly not know, that they need to know right now?*

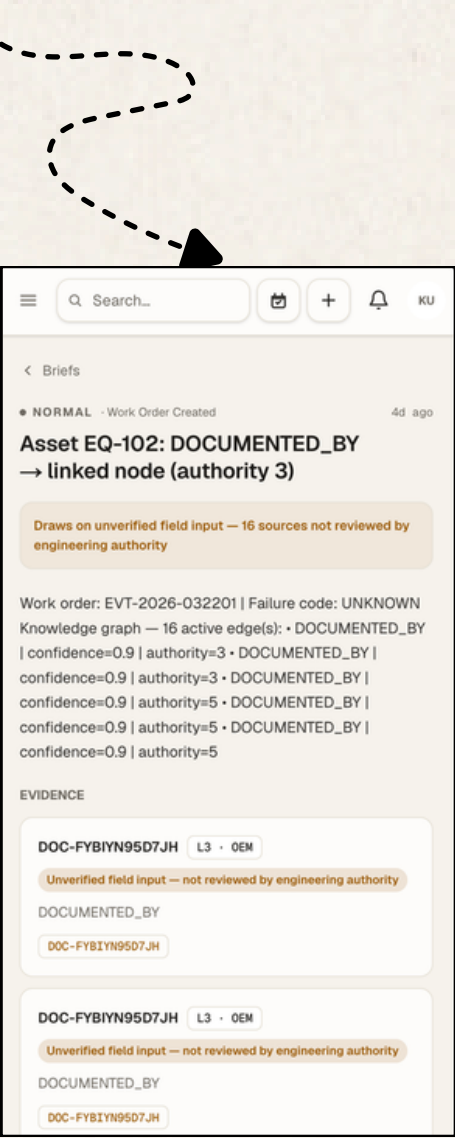
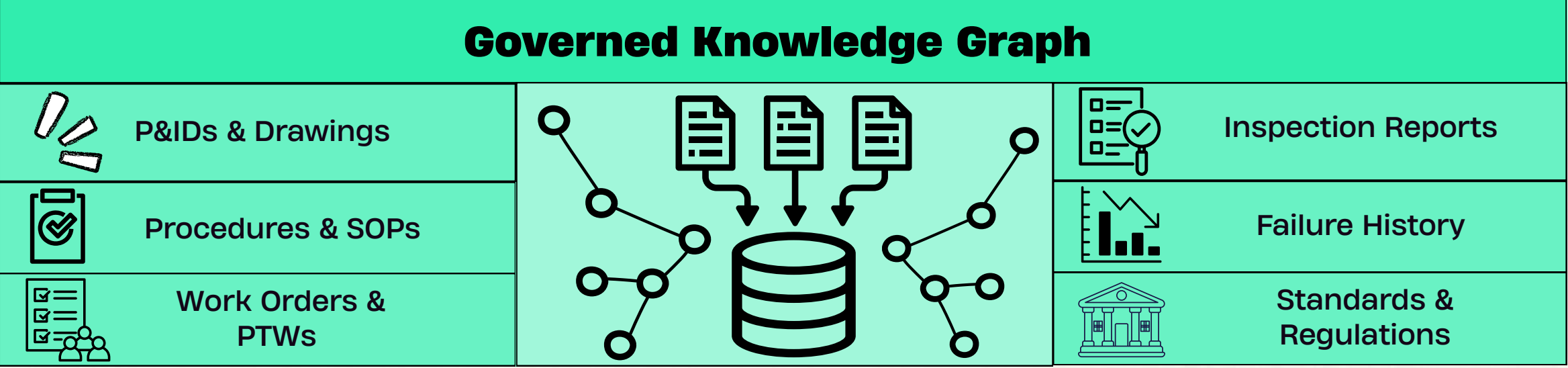
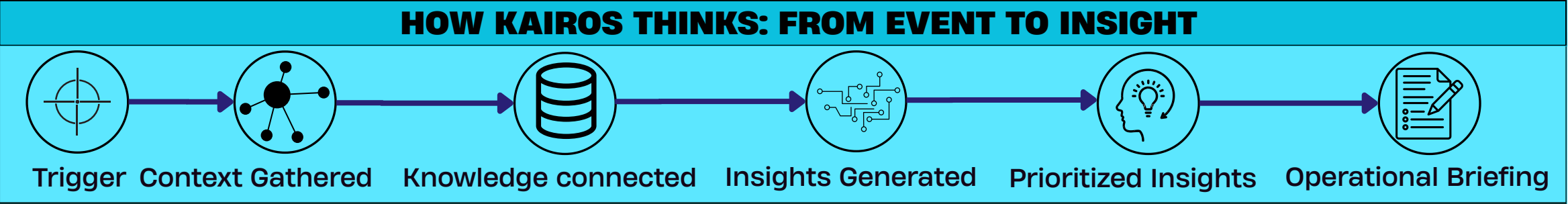
THE UNIT OF VALUE

The unit of value is not a search result. It is a contextually relevant operational briefing, delivered at the moment action is about to happen.

1.	Operational Event (Trigger)
	PTW Issued
	Work Order Created
	Shift Handover
	Alarm Acknowledged
	Equipment Tag-Out

2.	KAIROS ENGINE
	Understands the event and its context
	Connects with governed knowledge to find what matters
	Ranks by risk, relevance and recency
	Creates a concise, action-ready briefing

3.	Contextually Relevant Operational Briefing
	Relevant Risks & Hazards
	Applicable Procedures
	Similar Past Events & Lessons
	Compliance & Regulatory Updates
	Asset-Specific Instructions



Every other system waits to be asked. KAIROS decides what you need to know, and tells you first.

Modeled projection, built from published industry benchmarks. Not measured plant data. KAIROS has not yet been deployed at a live facility.

INDUSTRY BENCHMARKS USED IN THIS MODEL



Knowledge workers spend **20%** of the workweek searching for and gathering information

(McKinsey Global Institute, "The Social Economy," 2012)



A searchable, well-governed knowledge record can reduce time spent by up to **35%**

(McKinsey Global Institute, "The Social Economy," 2012)



Unplanned downtime costs the world's largest industrial companies nearly **\$1.4 trillion** globally per year, ~11% of annual

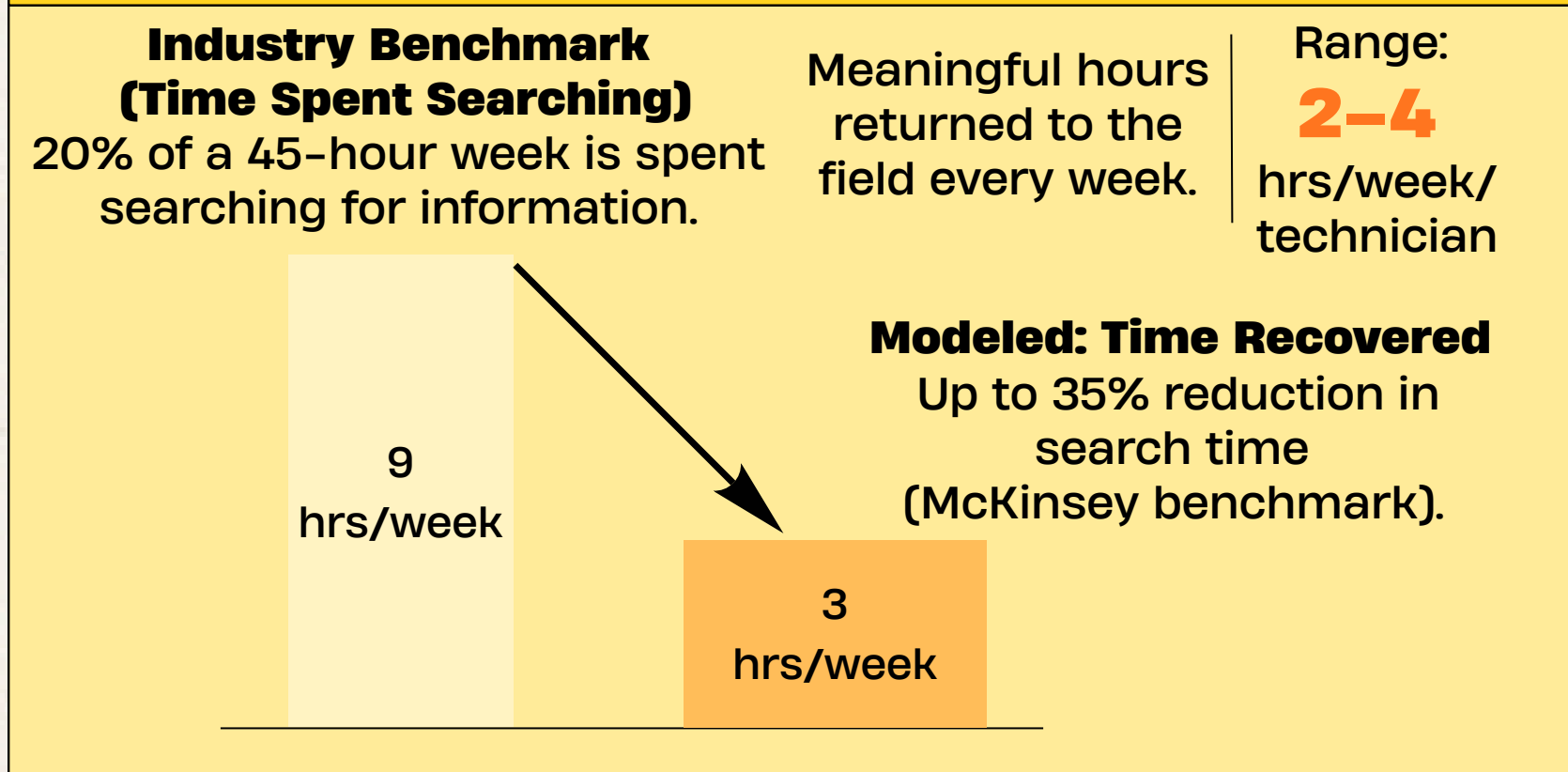
(Siemens, "True Cost of Downtime," 2024)



60% of surveyed engineers rate the loss of knowledge/information when an employee departs as "very" or "extremely" important

(IEEE, "Pulse of Engineering," 2019)

MODELED IMPACT: TIME RECOVERY PER TECHNICIAN



AUDIT PREPARATION ACCELERATION



KAIROS accelerates audit preparation by:

- Mapping compliance gaps to requirements
- Generating evidence packages
- Organizing artifacts with traceability & lineage

This is audit-preparation acceleration –not automated compliance.

HOW THE MODEL WORKS (AUDITABLE MATH)



Typical workweek 45 hours



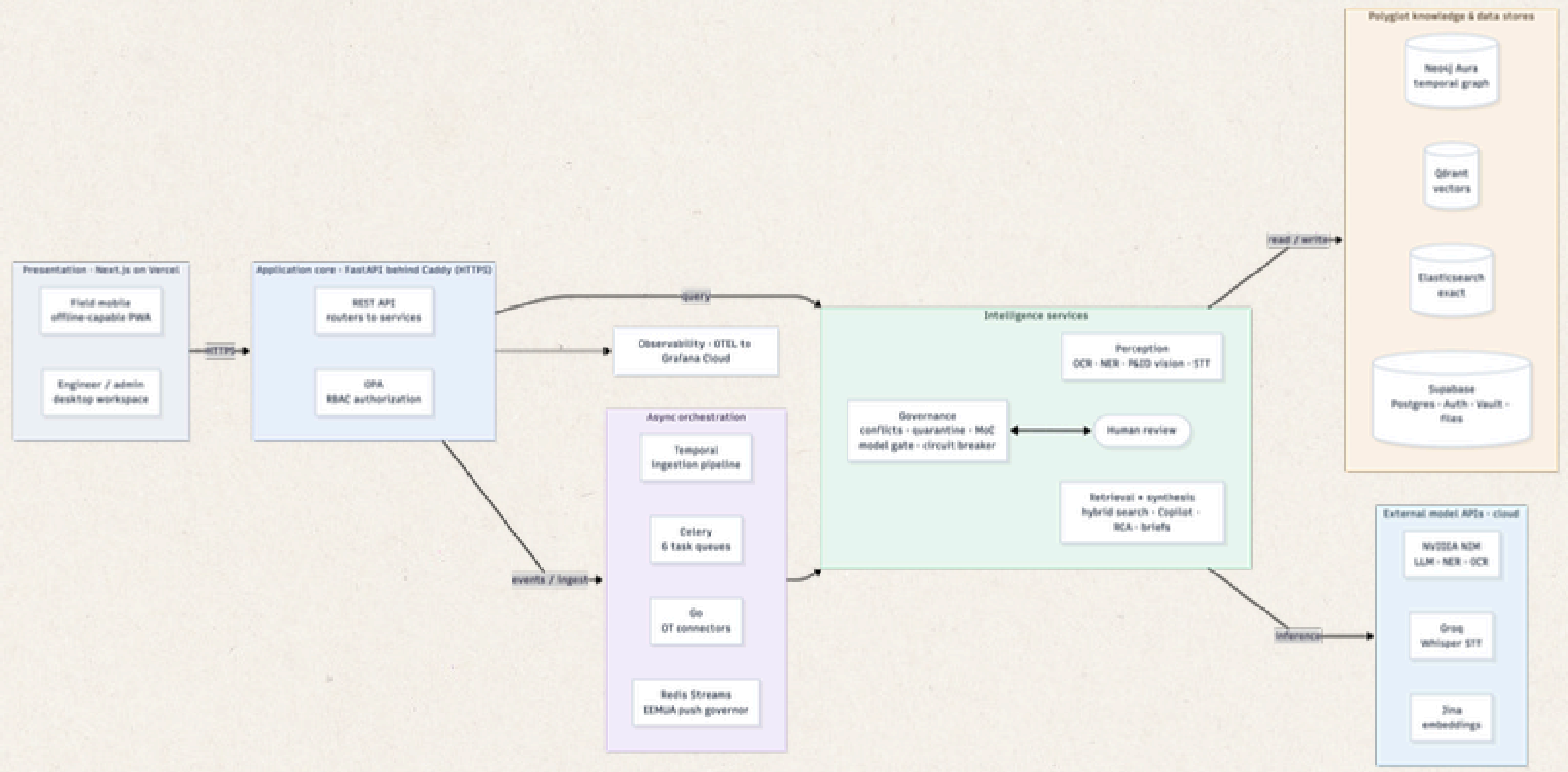
Time spent searching 20% of 45 hours = 9 hrs/week



Search-time reduction (McKinsey benchmark) upto 35%

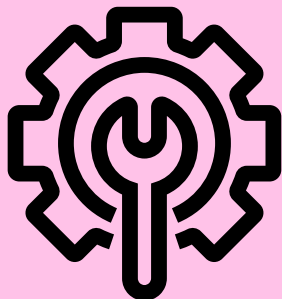


Modeled time recovered 9 hrs × 35% ≈ 3 hrs/week per technician

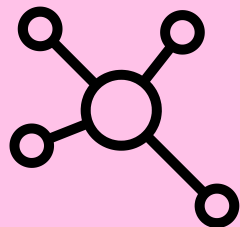


FLOW 1

WORK ORDER → PROACTIVE BRIEF



Work order opens for Pump P-101 (mechanical seal issue)

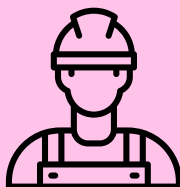


KAIROS queries the temporal graph:

- Prior failure history on this exact asset
- OEM manual guidance
- Any open MoC items on this equipment class



Brief assembled and pushed to the assigned technician's mobile app before they leave the workshop.



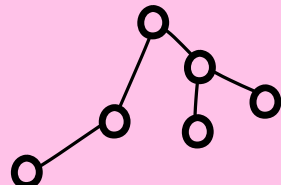
Technician arrives at the asset already knowing what failed before, under what conditions, and what the OEM manual specifies.

FLOW 2

PTW ISSUANCE → SAFETY BRIEF WITH SIGN-OFF



Permit-to-Work is issued for isolation work on a pressure vessel



KAIROS assemblies:

- Isolation topology (from P&ID graph)
- Any known deviations at this isolation point
- Applicable regulatory clause · Relevant near-miss history

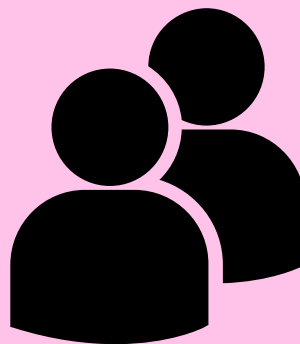


Issuing Engineer (Sign-off)



Shift Lead (Sign-off)

- Mandatory digital sign-off required from both before work proceeds.
- Cryptographically logged, creating an auditable record of what knowledge was available and confirmed received.

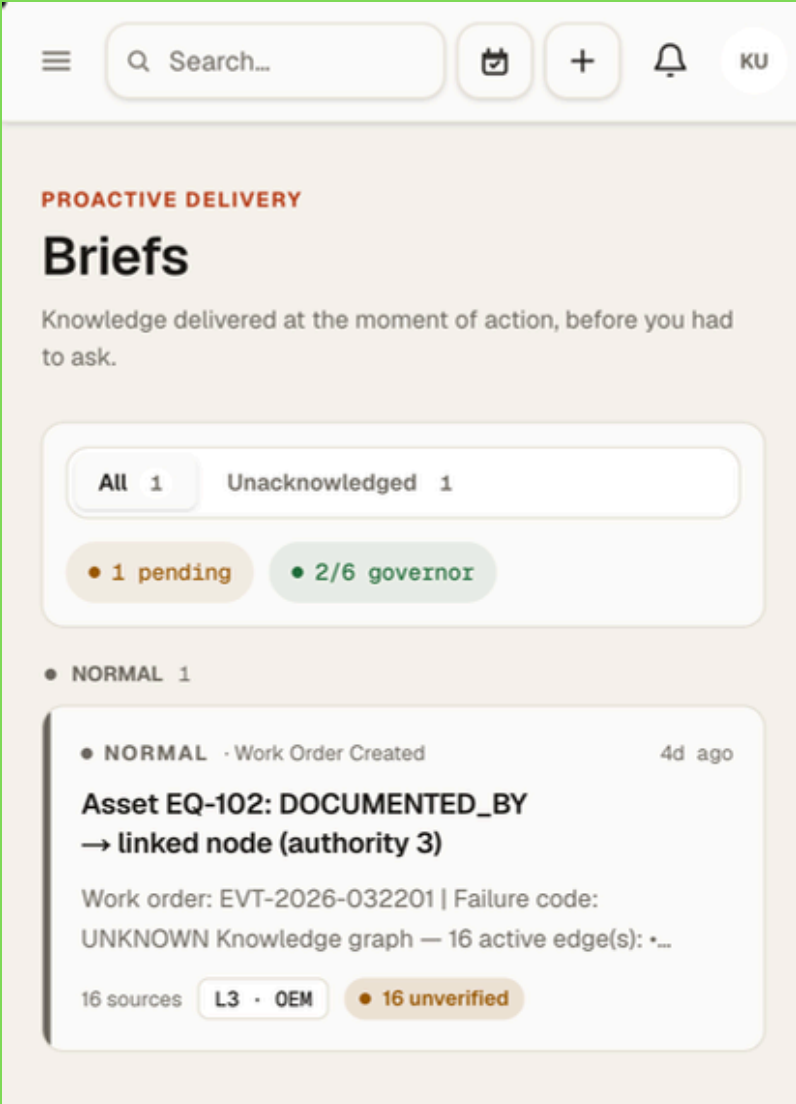


Brief delivered to both the issuing engineer and the shift lead.



Governance: Push volume is governed against the **EEMUA 191** alarm-management benchmark (Engineering Equipment and Materials Users' Association) to prevent **notification/alarm fatigue**, a well-established process-safety standard, not an invented ceiling. [VERIFIED, EEMUA 191 is a real, widely used industry alarm-management guideline]

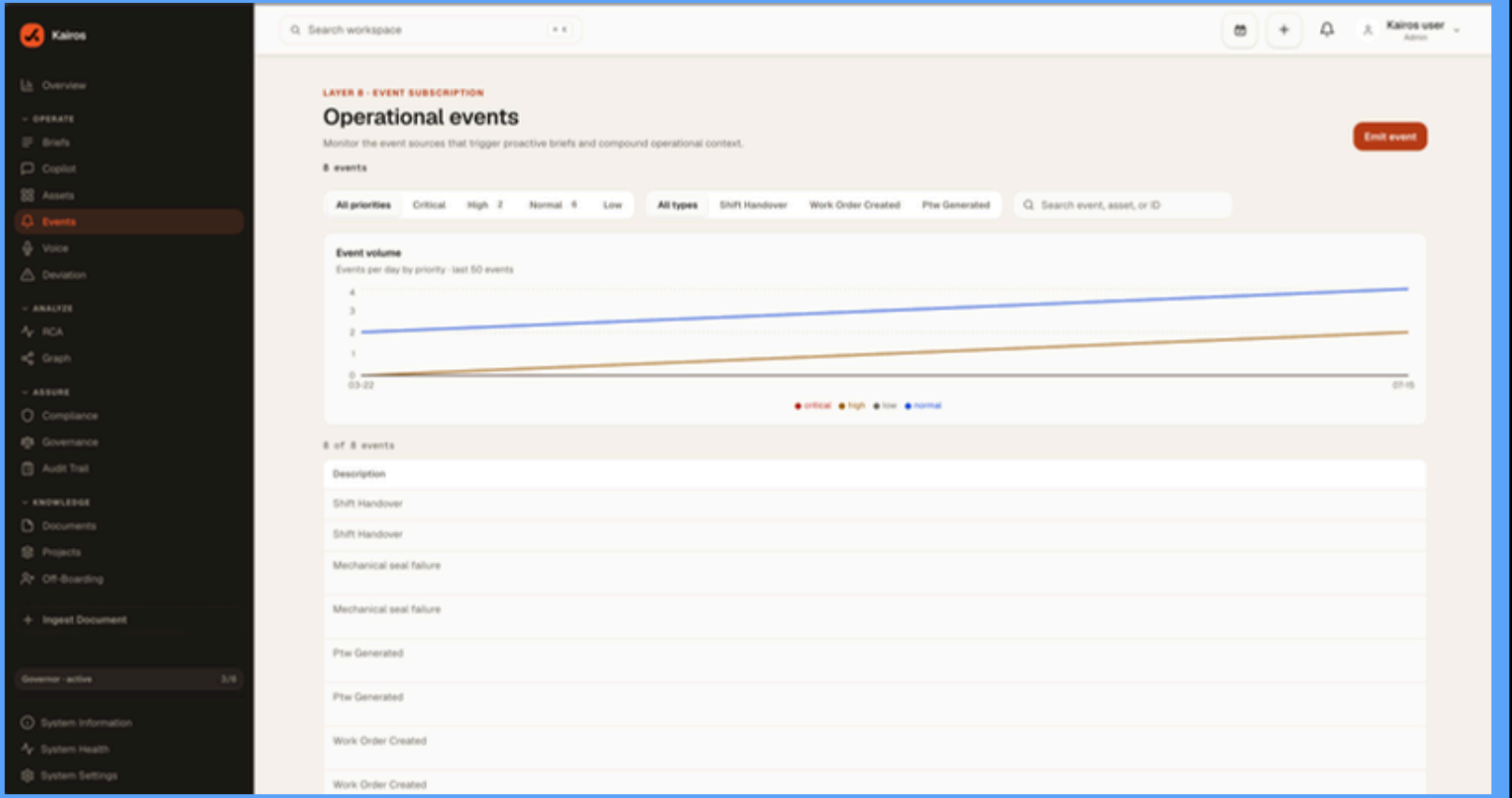
FIELD TECHNICIAN, MOBILE-FIRST
PROACTIVE BRIEF



- Single-handed operation, legible in direct sunlight, works offline with auto-sync
- Voice input for hands-free querying and elicitation responses
- Brief format built for reading while standing: key finding in the first two lines, detail below for those who need it
- One-tap acknowledgment, no menus, no query formulation required for an urgent question

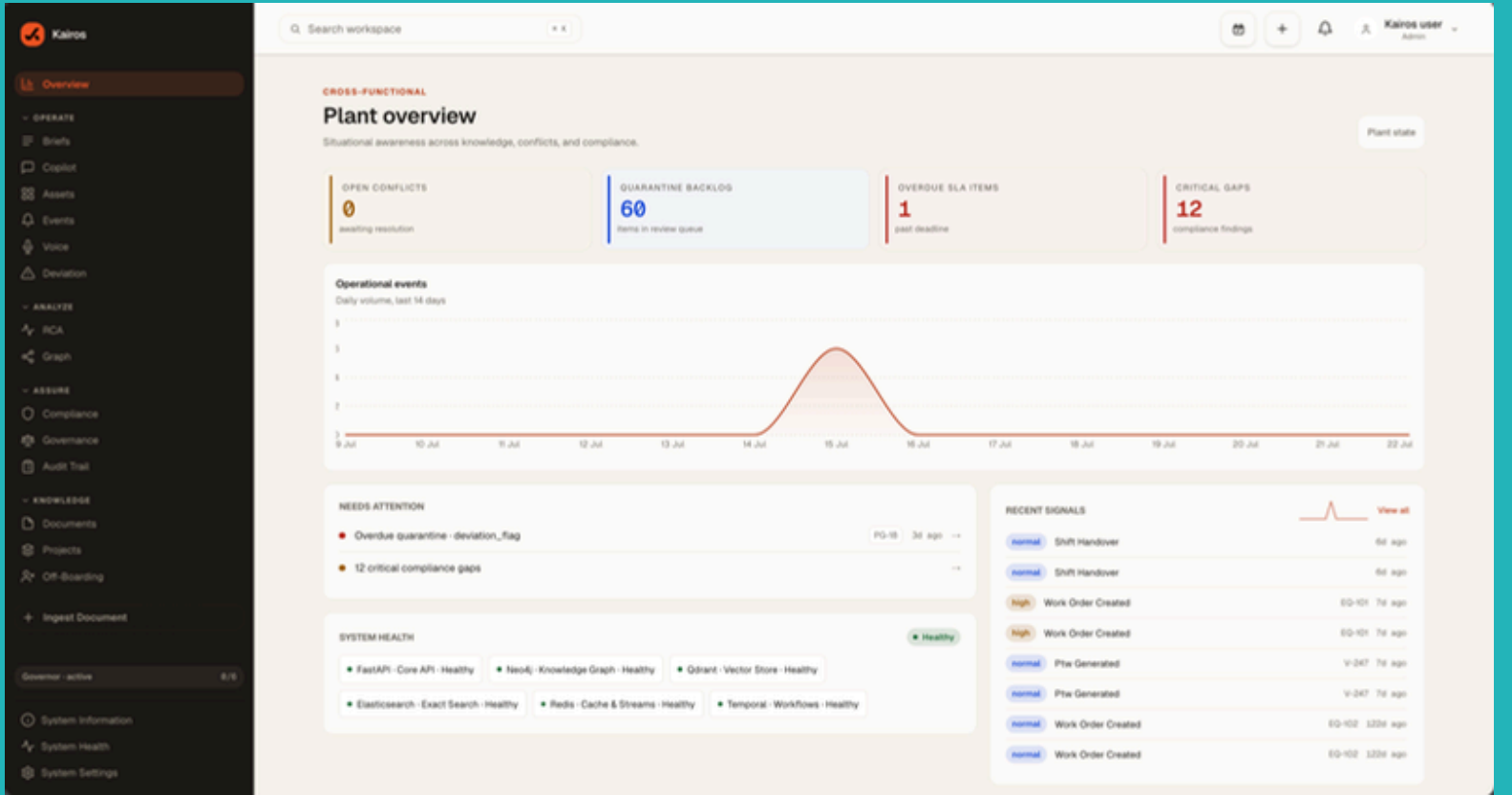
SHIFT LEAD,
HANDOVER
DASHBOARD & PTW
SIGN-OFF

- Structured shift handover briefing rather than informal verbal-only handoff
- PTW sign-off flow with full evidence lineage visible before acknowledgment



PLANT MANAGER /
QUALITY OFFICER,
COMPLIANCE COCKPIT

- Continuous compliance-gap dashboard mapped to the facility's applicable regulatory framework
- Knowledge-coverage heatmap (surfaces where the organization is dependent on a small number of non-participating experts)
- Evidence package assembly for audits



1.

Multi-script and multilingual perception

Language Support

- English
- Hindi
- Hinglish
- Mixed Script

Document Types

- Maintenance Forms
- Shift Logs
- Inspection Records



Captures knowledge from maintenance forms, shift logs, and inspection records across multiple language and scripts

2.

How it's handled ?

OCR NVIDIA NIMNemotron OCR-v2

=

Handles scanned, handwritten and multilingual documents

PDF PARSING PyMuPDF

=

Parses digital PDFs directly and efficiently

ENTITY EXTRACTION NVIDIA NIM Mistral-14B

=

JSON-structured prompting Extracts entities with high accuracy and context

OFFLINE FALLBACK

=

Ollama Llama3.1:8b - Local fallback for air-gapped or offline sites

3.

Active Learning Annotation Interface:



Low-Confidence Extractions Surface Inline

→

During normal search use.



One-Tap Correction by Operators

→

Quick & effortless fixes.



Facility-Specific Labeled Dataset Builds

→

Gets smarter over time.

4.	DPDP Act 2023 compliance for cross-site knowledge sharing	
	PII DETECTION	KAIROS detects PII in documents (names, shift IDs, personal attributes)
	AUTOMATED PII REDACTION	All personal identifiers and attributes are removed
	Facility-Specific Labeled Dataset Builds	Gets smarter over time.
	DPDP ACT 2023 COMPLIANT	Compliance ensured ahead of full enforcement

5.

P&ID topology extraction:

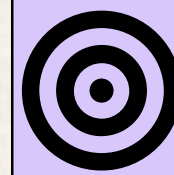


Engineering drawings are parsed for physical topology (which valve isolates which pump) via a cloud vision-language model today, with mandatory element-by-element engineer verification before anything becomes canonical.

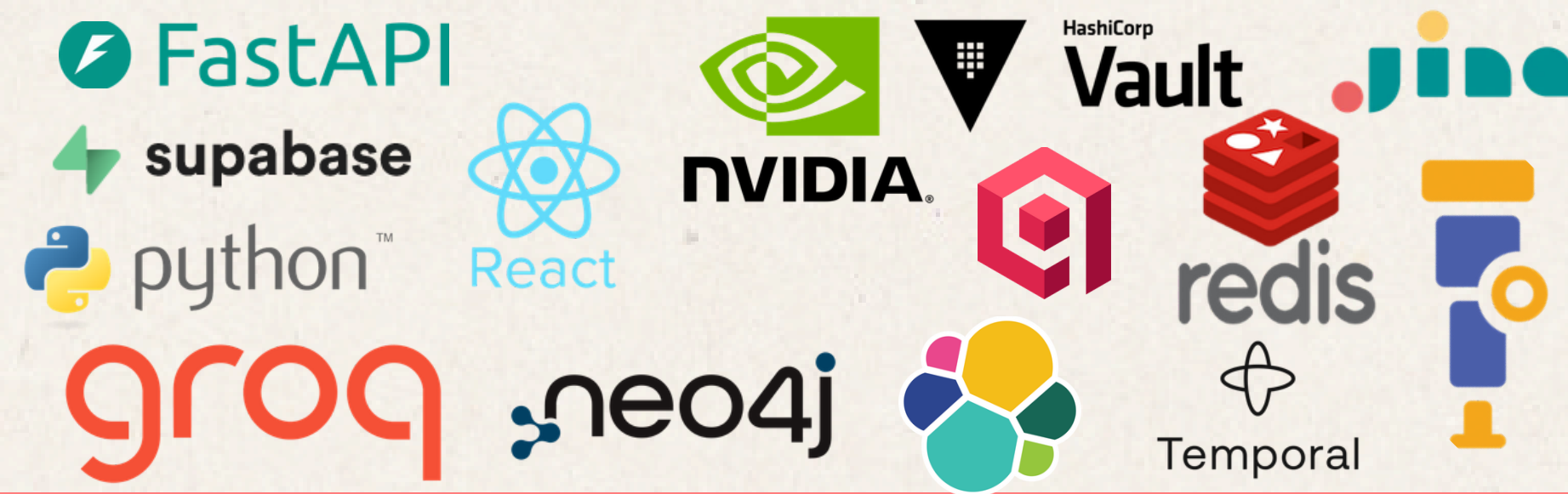


Purpose-built commercial drawing parsers report only **70-80%** automation rates.

DPDP ACT, 2023 – IMPLEMENTATION TIMELINE (Source: MeitY, GoI; DPDP Rules, 2025)

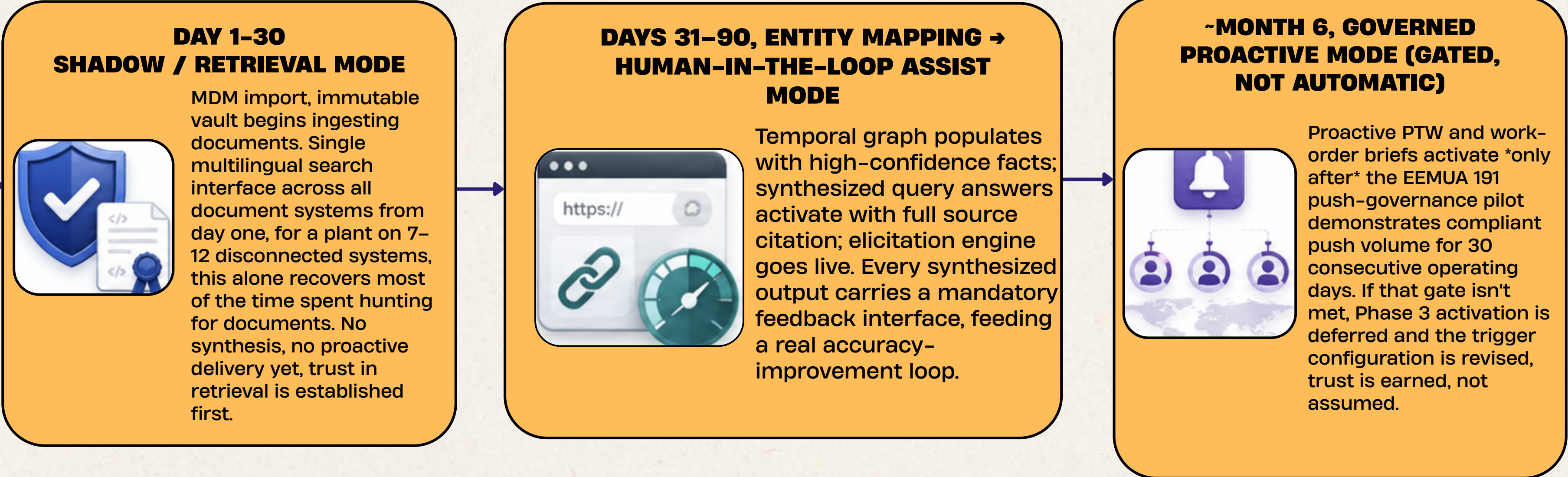
PHASE-1		PHASE-2		PHASE-3	
	Board Established Effective: 13 November 2025 COMPLETED		Consent Manager Framework Effective: 13 November 2026 WE ARE HERE (Mid-2026)		Full Compliance Obligations for All Data Fiduciaries Effective: 13 May 2027

TECH STACK	
Metric	Tool
Frontend	Next.js
Backend	FastAPI + Python
Machine Learning	PyTorch, Scikit-learn, LightGBM
Database	Neo4j (Knowledge Graph) + PostgreSQL + Qdrant (Vector Database)
AI	NVIDIA NIM (Nemotron-OCR-v2, Ministral-14B), Ollama (Llama 3.1:8B), Cloud Vision-Language Model
Deployment	Docker + NVIDIA GPU + Ollama



BUILT FULLY FOR THE DEMO	
	Document upload → OCR → NER → Neo4j graph → Qdrant vectors
	Hybrid retrieval + synthesis with source citations
	Work order → proactive brief → mobile delivery
	Supabase role-based auth (RBAC) and secure access
	Compliance gap detection against a configured regulatory framework

MOCKED / SIMULATED FOR THE DEMO	
	Custom YOLOv9 drawing parser (architecture shown, simplified pre-processed topology used)
	EEMUA 191 governor (rate-limiting logic implemented, DCS state feed mocked)
	Cross-site control plane (single-site for MVP, multi-site shown in architecture diagram only)



Control plane (central)

- enterprise-wide IAM & governance
- cross-site Ontology governance
- cross-site knowledge failure pattern detection

